Automated Food Expiry Detection and Shelf-Life Prediction Using OCR and Barcode-Based Identification

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Abstract

Food waste is a significant global problem causing economic inefficiencies, pollution, and improper usage of resources. Poor inventory management and ineffective monitoring of expiry dates cause unnecessary food waste in houses and small enterprises. This research paper presents a food expiry detection and reminders system using artificial intelligence (AI) for effective food inventory management. The suggested system combines optical character recognition (OCR) for food expiry date detection, barcode scanning for product identification, and food shelf life estimation for managing foods. For packaged food products, OCR technique is employed to detect manufacturing and expiry dates from food labels, while barcode scanning allows to identify the food product and manage the inventory. For the fresh foods without explicit expiry dates, the remaining shelf life is estimated based on the predefined food type dataset and user-defined storage conditions. Moreover, the proposed system provides centralized inventory management, flexible settings for shelf life, expiry detection, and automated reminders before food expiry. The system's performance is planned to be assessed based on accuracy, precision, recall, and F1-score for expiry date detection and proper error metrics for shelf life estimation. Preliminary performance goals are set for expiry-date identification, such as accuracy of 92%, precision of 94%, recall of 90%, and an F1-score of about 92%; these goals need to be verified experimentally via the use of an annotated dataset. The proposed framework is expected to contribute to improving food stock management as well as reducing food waste due to the application of automated expiry date detection.

Keywords - Artificial intelligence, barcode recognition, food inventory management, food waste reduction, optical character recognition (OCR), shelf-life estimation, expiry-date detection.